

SESSION 5

Ergonomics, Use Diagrams and Validation Plan

Turning product specifications into user interaction, safety criteria and test evidence.

Technical report fragment

Evidence-ready



Why this session matters

From “it should work” to “we can prove it works.”

Reports already have concept, requirements, materials, parts and BOM direction. Must explain how the product is used, where it touches the user or context, what could fail, and what evidence will prove performance.

Before

Concept
Requirements
Materials
Parts list

Today

Use sequence
Ergonomics
Risks
Validation plan

After

Final tests
Evidence
Production
report

Key idea

The report must not only describe the product. It must explain how the user interacts with it and how the team will test comfort, safety, stability, ease of use and performance.

Session 5 report structure

01 User-product interaction

How the user touches, wears, carries, opens, closes, adjusts or transforms the product.

02 Use sequence

Visual step-by-step explanation of the product in use.

03 Ergonomic analysis

Posture, grip, load, pressure, fit, movement and comfort.

04 Contact points + risks

Where the product touches the body, object, vehicle or environment, and what could fail.

05 Validation plan

Tests, success criteria and evidence needed.

06 Next steps

What must be measured, tested, redrawn or corrected before Session 6.

Use sequence: show the product in action

A use sequence makes the product understandable.

Must show the main interaction as a clear sequence. It can be sketches, photos, renders or simple diagrams.

1**Prepare**

What does the user do before using it?

2**Adjust / attach**

How is it placed, connected or fitted?

3**Use**

What is the main action or benefit?

4**Access / transform**

How does the user open, remove, fold or activate it?

5**Store / carry**

What happens after use?

6**Maintain**

How is it cleaned, charged, replaced or stored?

Ergonomic analysis: what to observe

Body fit

Size, adjustment, body area, posture and movement range.

Grip + access

How the hand reaches, holds, pulls, opens or detaches the product.

Load + weight

Where the weight goes and whether it causes fatigue or imbalance.

Pressure

Contact zones, discomfort, rubbing, sharp edges or tight areas.

What to include in the report

Area	Question	Output	Evidence
Posture	Does it force awkward movement?	Posture notes	Photo / diagram
Fit	Can different users adjust it?	Fit range	Test notes
Movement	Does it interfere with activity?	Movement check	Video
Comfort	Can it be used for a defined time?	Rating / comments	Survey

Critical contact points and safety risks

Mark the zones where the product touches or could fail.

This section should become useful evidence for both the technical report and the visual explanation of comfort, safety and trust.

Contact point	Possible risk	What to check	Evidence
Body	Pressure / discomfort	Fit, rubbing, posture	Annotated photos
Object / gear	Detachment / slipping	Grip, load, retention	Video test
Vehicle / board	Vibration / instability	Movement, balance, contact	Movement test
Mechanism	Pinch / breakage	Edges, force, repeated use	Risk checklist
Liquid / weather	Leak / damage	Water, sweat, cleaning	Basic test notes

Evidence needed: make it documentable

Photos

Use, contact points, scale, body fit, prototype detail.

Videos

Movement, access time, transformation, stability, load behavior.

Measurements

Time, dimensions, weight, load, distance, temperature, force.

Surveys

Comfort rating, difficulty, trust, understanding, purchase interest.

Diagrams

Use sequence, risk map, ergonomic map, test setup.

Tables

Test criteria, results, missing data and improvement plan.

What to submit for Session 5

- Cover / header with project name, team and main image
- User-product interaction: what the user does with the product
- Use sequence: visual step-by-step process
- Ergonomic analysis: posture, grip, weight, pressure, fit or movement
- Critical contact points and safety risks
- Validation plan with tests, success criteria and evidence
- Technical explanation for the final production report
- Visual/commercial explanation for the user-facing report
- Missing information and next steps

Suggested length: 8–12 pages. Use images, diagrams, tables, captions and short paragraphs.